Uniqueness of convex central configurations in the planar $(1+4)$-body problem

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Abstract

We study strictly convex central configurations of the planar Newtonian five-body problem with one dominant mass and four satellites of arbitrary positive masses. We prove that a single positive bound on the total satellite-to-primary mass ratio guarantees existence and uniqueness for every prescribed directed order of the five bodies on the convex hull, up to translations, rotations and positive scaling. The bound is independent of all satellite mass ratios, including ratios arbitrarily close to the boundary of the mass simplex. The constrained Hessian is nondegenerate on the six-dimensional shape space. We first prove uniqueness and nondegeneracy for the convex coorbital limit by an angular Hessian estimate. We then classify the possible collision limits as the satellite mass ratios vary and construct regularized local continuations for pairs, triples and pairs within triples. An exact factorization of the signed areas preserves strict convexity on independent parameter neighborhoods. These continuations connect every sufficiently small-mass configuration to the unique coorbital configuration and yield the common bound.

1 Introduction

The planar $(1+N)$-body problem describes the coorbital limit of central configurations with one dominant mass [2]. Its geometry and stability have been studied in [7, 11]; results for four satellites include the equal-mass classification [1, 5], stacked configurations [10], and mass symmetries [6]. For three satellites, Corbera, Cors and Llibre [3] and Corbera, Cors, Llibre and Moeckel [4] describe the bifurcations of relative equilibria. In particular, [4, Theorem 2] gives convex uniqueness for every fixed positive mass ratio and sufficiently small total satellite mass.

We prove the corresponding uniqueness statement for four satellites with a bound independent of their mass ratios. Write

$$m_0 = 1, \qquad m_i = \varepsilon\mu_i \quad (1 \le i \le 4), \qquad \mathcal{M} = \left\{ \mu \in \mathbb{R}^4_{>0} : \sum_{i=1}^{4} \mu_i = 1 \right\}. \tag{1.1}$$

Theorem 1.1. *There is $\delta_0 > 0$ such that, for every $\mu \in \mathcal{M}$, every $0 < \varepsilon < \delta_0$, and every directed counterclockwise hull order $(0, \ell_1, \ell_2, \ell_3, \ell_4)$, the masses (1.1) admit exactly one strictly convex central configuration of that order, modulo translations, rotations and positive scaling. Its constrained Hessian on the six-dimensional shape space is nondegenerate.*

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Here $(\ell_1, \dots, \ell_4)$ is a permutation of $(1, 2, 3, 4)$; reflections are not included in the equivalence relation. For unnormalized masses the condition is $\sum_{i=1}^4 m_i/m_0 < \delta_0$.

The proof first establishes convex coorbital uniqueness. Uniformity then requires control when some μ_i tend to zero and satellites coalesce. Following the cluster approach of [8], we classify these limits, regularize their equations, and connect every small-mass convex configuration to its coorbital limit. The signed-area estimates ensure that this continuation remains convex without imposing a relation between the mass parameter and the collision scales.

2 Convex coorbital configurations

Put $e(t) = (\cos t, \sin t)$ and $e^\perp(t) = (-\sin t, \cos t)$. In the standard hull order, the coorbital configurations are parametrized by

$$\theta_1 = 0, \qquad (\theta_2, \theta_3, \theta_4) \in \mathcal{D} = \{0 < \theta_2 < \theta_3 < \theta_4 < \pi\}. \tag{2.1}$$

Indeed, the origin and four distinct unit-circle points form a strictly convex pentagon exactly when the points lie in an open semicircle.

We use V and A as in [4, Section 2]. For $0 < t < \pi$, set

$$f(t) = \sin t\big((2\sin(t/2))^{-3} - 1\big), \qquad v(t) = \frac{1}{2\sin(t/2)} + 1 - \cos t, \qquad \kappa = -f' = v''. \tag{2.2}$$

Extend f oddly. With $r_{ij}(\theta) = |e(\theta_i) - e(\theta_j)|$, the equations are

$$E_i = \sum_{j \neq i} \mu_j f(\theta_j - \theta_i) = 0, \qquad V = \sum_{i<j} \mu_i \mu_j \left(\frac{1}{r_{ij}(\theta)} + \frac{r_{ij}(\theta)^2}{2} \right). \tag{2.3}$$

Thus $E = A(\theta)\mu$, where $A_{ii} = 0$ and $A_{ij} = f(\theta_j - \theta_i)$, and $\partial_{\theta_i} V = \mu_i E_i$. Write $h = \pi/3$. Differentiation gives

$$\begin{aligned} \kappa(t) &= \frac{2 - \sin^2(t/2)}{8\sin^3(t/2)} + \cos t, \\ \kappa'(t) &= -\frac{\cos(t/2)}{16\sin^4(t/2)}\big(6 - \sin^2(t/2) + 32\sin^5(t/2)\big) < 0. \end{aligned} \tag{2.4}$$

We shall use $f(h) = f(\pi) = 0$, $f > 0$ on $(0, h)$, $f < 0$ on (h, π), $\kappa(h) = 9/4$, $f \geq -1$, and $tf(t) \to \infty$ as $t \downarrow 0$.

2.1 An angular Hessian estimate

Lemma 2.1. *Suppose $2 \leq N \leq 4$ ordered points in an open semicircle carry positive masses w_i and satisfy the angular equations. Put*

$$\ell(t) = \begin{cases} f(t)/(h - t), & t < h, \\ 0, & t = h, \\ f(t)/(\pi - t), & t > h, \end{cases} \qquad b(t) = \kappa(t) - \ell(t).$$

Then $b(t) > 0$, and, with $t_{ij} = \theta_j - \theta_i$ for $i < j$,

$$\sum_{i<j} w_i w_j \kappa(t_{ij})(u_j - u_i)^2 \geq \sum_{i<j} w_i w_j b(t_{ij})(u_j - u_i)^2. \tag{2.5}$$

In particular, the angular Hessian has kernel $\mathbb{R}\mathbf{1}$.

Proof. Since κ decreases strictly,

$$f(t) = \int_t^h \kappa(s)\,\mathrm{d}s < (h-t)\kappa(t) \quad (t<h), \qquad f(t) = \int_t^\pi \kappa(s)\,\mathrm{d}s < (\pi-t)\kappa(t) \quad (t>h).$$

Together with $\kappa(h) > 0$, this proves $b > 0$.

Orient an edge $i < j$ rightwards if $t_{ij} < h$ and leftwards if $t_{ij} > h$, with flow $w_i w_j |f(t_{ij})|$; omit zero edges. The equations express balance at every vertex, so the flow decomposes into positive flows on simple directed cycles. Such a cycle has at most four edges. Two leftward edges would have total length $> 2h$, whereas the remaining rightward edges have total length $< 2h$. Hence there is one leftward edge and $k = 2$ or 3 rightward edges.

Let the rightward lengths and increments be t_r, u_r, with $T = \sum_r t_r \in (h, \pi)$. The contribution of this cycle to the form with coefficient ℓ is its flow times

$$\sum_{r=1}^{k} \frac{u_r^2}{h - t_r} - \frac{(\sum_r u_r)^2}{\pi - T} \geq \frac{(\sum_r u_r)^2}{kh - T} - \frac{(\sum_r u_r)^2}{\pi - T} \geq 0,$$

because $0 < kh - T \leq \pi - T$. Summing over cycles proves (2.5). Its right side is positive for every nonconstant vector, while rotation invariance supplies the constant kernel. □

Theorem 2.2. *For each $\mu \in \mathcal{M}$, equations* (2.3) *have exactly one root $\theta^0(\mu)$ in* (2.1)*. The Hessian D^2V is positive definite on the slice $\theta_1 = 0$, and θ^0 depends smoothly on μ.*

Proof. The Hessian form is the left side of (2.5), so the lemma gives positivity. We prove properness of the projection of the zero set to $\mathcal{M}$. Let a, b, c be the consecutive gaps and $L = a + b + c$. Summing the mass-weighted equations over the first k labels gives

$$\sum_{i \leq k < j} \mu_i \mu_j f(\theta_j - \theta_i) = 0, \qquad k = 1, 2, 3. \tag{2.6}$$

The signs of f imply $a, b, c < h < L$. For μ in a compact subset of $\mathcal{M}$, let $\alpha > 0$ be a common lower bound on the masses. Pairing terms in $\sum_i \mu_i \theta_i E_i = 0$ yields

$$\sum_{i<j} \mu_i \mu_j t_{ij} f(t_{ij}) = 0.$$

The negative terms sum to at least $-\pi/2$, so every consecutive gap d satisfies $\alpha^2 d f(d) \leq \pi/2$. Collisions are therefore excluded. If $L \to \pi$, the gaps tend to h, and the first equation tends to $\mu_3 f(2h) = 0$, a contradiction. This proves properness.

Since $\sum_i \mu_i E_i = 0$, the last three equations recover the first. Their Jacobian in $(\theta_2, \theta_3, \theta_4)$ is invertible by the Hessian estimate. The mass projection is thus a proper local diffeomorphism. Its nonempty image is open and closed in the connected simplex, and it is a covering with finitely many sheets; cf. [4, proof of Proposition 3]. By the classification in [1, main theorem], exactly one equal-mass configuration has its satellites in an open semicircle, and its occupied gaps are symmetric under reversal. The equal-mass fiber has one point, so every fiber has one point. Local inversion gives smoothness. □

2.2 Zero-mass satellites at distinct positions

The divided equations $E_i = 0$ are retained when $\mu_i = 0$. We first record the two-source probe estimate used below; cf. [6, Lemma 2].

Lemma 2.3. *Let $p, q \geq 0$, $p + q = 1$. If $0, h, x$ are distinct, have span less than π, and $pf(x) + qf(x-h) = 0$, then*

$$p\kappa(|x|) + q\kappa(|x-h|) > 0.$$

Proof. For $pq > 0$, put $g(x) = pf(x) + qf(x-h)$. On $(0,h)$, $g' < 0$ because $\kappa > 0$ there. On (h,π), $\kappa' < 0$ gives $g'' > 0$, while $g(\pi) = qf(\pi - h) < 0$; at a root, $g'(x) < g(\pi)/(\pi - x) < 0$. Reflection $x \mapsto h - x$ treats $(h - \pi, 0)$. If $pq = 0$, the claimed value is $\kappa(h) = 9/4$. □

Proposition 2.4. *Suppose $\mu_i \geq 0$, $\sum_i \mu_i = 1$, four distinct angles lie in an open semicircle, and all E_i vanish. Then $J = D_\theta E$ has kernel $\mathbb{R}\mathbf{1}$. Deleting a row p with $\mu_p > 0$ and any column gives an invertible 3×3 matrix. Its inverse is uniformly bounded on closed sets of such roots with consecutive gaps bounded below, span bounded away from π, and $\mu_p \geq 1/4$.*

Proof. Let P, Z be the positive and zero supports. One positive mass would allow probes only at the two angles $\theta_p \pm h$, so $|P| \geq 2$. Lemma 2.1 gives the constant kernel on P. For $k \in Z$, set $D_k = \sum_{j \in P} \mu_j \kappa(|\theta_j - \theta_k|)$. If $|P| = 2$, the support separation is h, and Lemma 2.3 gives $D_k > 0$.

If $|P| = 3$, the nonzero skew matrix A_{PP} has rank two. Since $A\mu = 0$, the 4×4 skew matrix A also has rank two. Its two-dimensional kernel cannot consist only of vectors vanishing at the unique label $k \in Z$, since restriction would inject it into $\ker A_{PP}$. Choose $\nu \in \ker A$ with $\nu_k > 0$. For small $s > 0$, $\mu + s\nu$ is positive and solves the same angular equations. Apply (2.5) to the kth coordinate vector, divide by $s\nu_k$, and let $s \downarrow 0$:

$$D_k \geq \sum_{j \in P} \mu_j b(|\theta_j - \theta_k|) > 0.$$

In support order,

$$J = \begin{pmatrix} J_{PP} & 0 \\ J_{ZP} & \operatorname{diag}(D_k : k \in Z) \end{pmatrix}.$$

Every kernel vector is therefore constant. Also $\mu^T J = 0$ and $J\mathbf{1} = 0$. If one coordinate of a vector is zero and the retained rows annihilate it, the identity $\mu^T J = 0$ recovers the omitted row; the vector is constant and hence zero. The inverse bound follows from compactness and continuity. □

3 Full equations and local continuation

We adopt the centered normalization and notation of [4, Section 2]:

$$\sum_{j \neq i} m_j \frac{x_j - x_i}{r_{ij}^3} = -x_i, \qquad \sum_{i=0}^{4} m_i x_i = 0, \qquad r_{ij} = |x_i - x_j|, \tag{3.1}$$

and

$$U = \sum_{i<j} \frac{m_i m_j}{r_{ij}}, \qquad I = \sum_{i=0}^{4} m_i |x_i|^2, \qquad G = U + \tfrac{1}{2} I. \tag{3.2}$$

Thus $x_i = r_i e(\theta_i)$ denotes a centered polar coordinate. For the collision analysis use the primary-relative coordinates

$$q_i = x_i - x_0 = \rho_i e(\varphi_i), \qquad X = \sum_{i=1}^{4} \mu_i q_i, \qquad Y = \sum_{i=1}^{4} \mu_i q_i / \rho_i^3, \qquad c_\varepsilon = (1+\varepsilon)^{-1}.$$

Then $q_0 = 0$, $x_0 = -\varepsilon c_\varepsilon X$, $x_i = q_i - \varepsilon c_\varepsilon X$, and $r_{ij} = |q_i - q_j|$. At $\varepsilon = 0$ the centered and relative coordinates coincide. Below, sums over satellite labels run from 1 to 4.

Subtracting the primary equation gives

$$F_i = F_0(q_i) + \varepsilon \sum_{j \neq i} \mu_j \frac{q_j - q_i}{r_{ij}^3} - \varepsilon Y = 0, \qquad F_0(q) = q - \frac{q}{|q|^3}. \tag{3.3}$$

These differences recover (3.1), since all acceleration residuals are then equal and their mass-weighted sum is zero. In relative coordinates,

$$\Phi = \frac{G}{\varepsilon} = \sum_i \mu_i \left(\rho_i^{-1} + \frac{\rho_i^2}{2} \right) + \varepsilon \sum_{i<j} \frac{\mu_i \mu_j}{r_{ij}} - \frac{\varepsilon c_\varepsilon}{2} |X|^2. \tag{3.4}$$

Indeed $I = \varepsilon \sum_i \mu_i \rho_i^2 - \varepsilon^2 c_\varepsilon |X|^2$. For positive masses, $\nabla_q \Phi = (C \otimes I_2)F$, where $C = \operatorname{diag}(\mu_i) - \varepsilon c_\varepsilon \mu \mu^T$ is positive definite for $\varepsilon > -1$. Thus $F = 0$ is equivalent to $\nabla \Phi = 0$.

Put $g(\rho) = \rho - \rho^{-2}$ and $\Delta_{ij} = \varphi_j - \varphi_i$. The mass-divided radial and angular residuals are

$$R_i = g(\rho_i) - \varepsilon \sum_{j \neq i} \mu_j \frac{\rho_i - \rho_j \cos \Delta_{ij}}{r_{ij}^3} - \varepsilon c_\varepsilon X \cdot e(\varphi_i), \tag{3.5}$$

$$T_i = \sum_{j \neq i} \mu_j \rho_i \rho_j \sin \Delta_{ij} (r_{ij}^{-3} - c_\varepsilon). \tag{3.6}$$

They satisfy, also away from solutions,

$$\partial_{\rho_i} \Phi = \mu_i R_i, \qquad \partial_{\varphi_i} \Phi = \varepsilon \mu_i T_i, \qquad \sum_i \mu_i T_i = 0. \tag{3.7}$$

For $\varepsilon > 0$ and positive masses, retain four radial rows on a rotation slice and omit one angular row. The resulting seven equations are equivalent to (3.1). Their explicit divided rows remain meaningful at zero masses.

For estimates we also use the tangential projection of (3.3):

$$Q_i = \sum_{j \neq i} \mu_j \rho_j \sin \Delta_{ij} (r_{ij}^{-3} - \rho_j^{-3}) = 0. \tag{3.8}$$

At solutions, $\sum_i \mu_i F_i = X - (1 + \varepsilon) Y = 0$, so $Y = c_\varepsilon X$ and $T_i = \rho_i Q_i$.

Lemma 3.1. *For positive masses and $\varepsilon > 0$, the seven-equation Jacobian on a rotation slice is invertible if and only if the constrained shape Hessian is nondegenerate. Invertible coordinate changes and row operations preserve this equivalence.*

Proof. When the omitted row and fixed angle have the same label, (3.7) identifies the slice Hessian of Φ with the residual Jacobian multiplied by a positive diagonal matrix. Any other positive-mass row can be omitted using $\sum_i \mu_i T_i = 0$. At a root, an invertible coordinate map Ψ and row map L give $D(L \nabla \Phi \circ \Psi) = L (D^2 \Phi) D\Psi$.

Let $M = \operatorname{diag}(m_0 I_2, \dots, m_4 I_2)$, $E = \{v : \sum_i m_i v_i = 0\}$, and $T = \{v \in E : \langle Mx, v \rangle = 0\}$. The constrained Hessian is $D^2 G|_T$ modulo the rotation direction; see [9]. Homogeneity gives

$$D^2 G(x) x = 3Mx, \qquad E = \mathbb{R}x \oplus T,$$

an orthogonal splitting for $D^2 G$ with radial value $3I > 0$. Relative coordinates give a linear isomorphism onto E, and $G = \varepsilon \Phi$. Passing to a transverse rotation slice and removing this nonzero radial direction leaves the six-dimensional shape Hessian. □

We repeatedly use the following compact form of the implicit function theorem.

Lemma 3.2. *Let $F(y; s, \beta)$ be smooth near $\{(y^0(\beta), 0, \beta) : \beta \in K\}$, where K is compact, $F(y^0(\beta); 0, \beta) = 0$, and $D_y F$ is invertible. If the base roots and inverse matrices depend continuously on β, then the local roots exist uniquely on uniform independent parameter neighborhoods, with a common root radius and inverse bound.*

Proof. Apply the implicit function theorem locally and take a finite cover of the compact base graph. □

Proposition 3.3. *For each $K \Subset \mathcal{M}$, the coorbital graph $(1,1,1,1,\theta_2^0(\mu),\theta_3^0(\mu),\theta_4^0(\mu))$ has a unique nearby smooth solution branch for sufficiently small $|\varepsilon|$, uniformly for μ near K. For $\varepsilon > 0$, it is strictly convex in the same order and has nondegenerate shape Hessian.*

Proof. This is the continuation of [4, Proposition 1 and its proof]. At $\varepsilon = 0$, the radial block is $3I_4$, its angular columns vanish, and the reduced angular block is invertible by Theorem 2.2. Lemma 3.2 gives compact uniformity, the limiting signed areas preserve strict convexity, and Lemma 3.1 gives nondegeneracy. □

For the divided differences below we write $s(t) = \sin t/t$, with its smooth value $s(0) = 1$.

4 Uniform geometry and collision limits

Throughout this section, $\varepsilon \to 0$ along actual multiplier-one solutions with arbitrary positive satellite ratios. After passage to a subsequence, a complete cluster consists of all labels with the same limiting position. Constants may depend on the selected sequence, but not on its index.

4.1 Uniform bounds

Proposition 4.1. *Every configuration satisfies* $\operatorname{diam}(x) \le 4(1+\varepsilon)^{1/3}$. *For each $E > 0$ there is $b(E) > 0$ such that $\rho_i \ge b(E)$ whenever $0 < \varepsilon \le E$. Uniformly over all positive ratios and all configurations,* $\max_i |\rho_i - 1| \to 0$ *as* $\varepsilon \downarrow 0$.

For strictly convex configurations in standard order, there are $\eta_0, \varepsilon_0 > 0$, independent of the ratios, such that

$$\eta_0 \le \varphi_4 \le \pi - \eta_0 \qquad (0 < \varepsilon < \varepsilon_0). \tag{4.1}$$

Each consecutive angular gap has limit superior at most h.

Proof. Project onto a diameter direction and split the five labels across their largest consecutive projection gap $g \ge \operatorname{diam}(x)/4$. If the two groups have masses M_A, M_B, their projected centers are separated by

$$g \le \Delta = \left(M_A^{-1} + M_B^{-1}\right) \sum_{i\in A, j\in B} m_i m_j \frac{(x_j - x_i)\cdot e}{|x_j - x_i|^3} \le \frac{1+\varepsilon}{g^2}.$$

This proves the diameter bound.

If satellites approached the primary with $\varepsilon \le E$, take the complete limiting cluster C containing it. Let $d \to 0$ be its diameter, s its satellite mass, b its center, and I_C, U_C its internal inertia and potential. Then $U_C \ge s/d$ and $I_C \le sd^2$. The exterior acceleration A has a uniformly bounded derivative near C. Summing the cluster equations against $m_i(x_i - b)$ gives

$$U_C = I_C + \sum_{i\in C} m_i (x_i - b)\cdot[A(x_i) - A(b)] \le (1+L) I_C.$$

Cancellation of s contradicts $d \to 0$ and proves the uniform lower radius bound.

Now let C be a complete satellite cluster with nonzero limit q^0. Average (3.3) with the current weights $\mu_i / \sum_{j\in C} \mu_j$. Internal forces cancel, and all exterior distances stay positive. Since Y is bounded, the average is $F_0(q^0) + o(1) = O(\varepsilon)$. Thus $|q^0| = 1$; a violating sequence proves uniform convergence of all radii.

Strict convexity gives $0 = \varphi_1 < \cdots < \varphi_4 < \pi$. Summing the first k weighted angular equations gives

$$\sum_{i \le k < j} \mu_i \mu_j \rho_i \rho_j \sin(\varphi_j - \varphi_i)(r_{ij}^{-3} - c_\varepsilon) = 0. \tag{4.2}$$

A gap exceeding h by a fixed amount makes every term negative for small ε. If $\varphi_4 \to \pi = 3h$, all three gaps tend to h. But the skew matrix

$$B_{ij} = \rho_i \rho_j \sin(\varphi_j - \varphi_i)(r_{ij}^{-3} - c_\varepsilon), \qquad B_{ii} = 0,$$

satisfies $B\mu = 0$, whereas its limiting determinant is $f(2h)^4 > 0$. If $\varphi_4 \to 0$, all satellite distances vanish and every term of (4.2) is positive. These contradictions give (4.1). □

We shall use

$$r_{ij}^2 = (\rho_j - \rho_i)^2 + 4\rho_i \rho_j \sin^2 \frac{\varphi_j - \varphi_i}{2}, \tag{4.3}$$

and the strict support inequality for any adjacent hull edge $a \to b$ and any other vertex j:

$$\det(q_b - q_a, q_j - q_a) > 0. \tag{4.4}$$

4.2 A complete pair

Proposition 4.2. *Let $\{a, b\}$ be a complete pair, $a < b$, and put $d = |q_b - q_a|$, $\nu = \mu_a + \mu_b$, $\sigma = 1 - \nu$, $\delta = \varphi_b - \varphi_a$, and $\Delta\rho = \rho_b - \rho_a$. The pair is adjacent and*

$$\nu \to 0, \qquad \delta/d \to 1, \qquad |\Delta\rho| \le C\varepsilon\sigma d, \qquad \nu/d^3 \to D > 0. \tag{4.5}$$

The two exterior limits are distinct and separated by h. In the gauge placing them at $0, h$, with limiting masses $p, 1-p$ and pair angle ψ,

$$pf(\psi) + (1-p)f(\psi - h) = 0, \qquad D = p\kappa(|\psi|) + (1-p)\kappa(|\psi - h|).$$

Two complete pairs cannot coexist.

Proof. Equal-angle classes are consecutive blocks. The exterior fields

$$P(q) = \sum_{j \notin \{a,b\}} \mu_j \frac{q_j - q}{|q_j - q|^3}, \qquad H(\rho, v) = \sum_{j \notin \{a,b\}} \mu_j \rho_j \sin(\varphi_j - v)(|\rho e(v) - q_j|^{-3} - \rho_j^{-3})$$

have first derivatives $O(\sigma)$ near the pair. Subtraction of (3.8) gives

$$H(\rho_b, \varphi_b) - H(\rho_a, \varphi_a) = \sin\delta \left(\frac{\mu_a \rho_a + \mu_b \rho_b}{d^3} - \frac{\mu_a}{\rho_a^2} - \frac{\mu_b}{\rho_b^2} \right) > 0. \tag{4.6}$$

Since $\delta \le Cd$, this implies

$$\delta\nu/d^3 \le C\sigma d. \tag{4.7}$$

The vector difference is

$$F_0(q_b) - F_0(q_a) + \varepsilon[P(q_b) - P(q_a)] = \lambda(q_b - q_a), \qquad \lambda = \varepsilon\nu/d^3. \tag{4.8}$$

Its integral derivative bounds λ. Since $DF_0 \to \mathrm{diag}(3, 0)$ in the radial–tangential basis, a limiting unit pair direction is radial with $\lambda \to 3$, or positively tangential with $\lambda \to 0$.

In a radial subsequence, $\delta = o(d)$ and $|\Delta\rho| \sim d$. The support inequality expands as

$$\det(q_b - q_a, q_j - q_a) = \Delta\rho\, \rho_j \sin(\varphi_j - \varphi_a) + o(d).$$

An interior pair would require opposite signs of $\Delta\rho$. For a left endpoint pair it requires $\Delta\rho > 0$, whereas on its polar rectangle

$$H_\rho = -3 \sum_{j \notin \{a,b\}} \mu_j \rho_j \sin(\varphi_j - v)[\rho - \rho_j \cos(\varphi_j - v)]|\rho e(v) - q_j|^{-5} \le -\sigma/C.$$

Thus the left side of (4.6) is at most $-\sigma|\Delta\rho|/C + C\sigma\delta < 0$. At the right endpoint both radial signs reverse, giving the same contradiction.

Consequently $\delta/d \to 1$, $\Delta\rho/d \to 0$, and (4.7) yields $\nu/d^3 \le C\sigma$, hence $\nu \to 0$. Projecting (4.8) onto the midpoint ray gives exactly

$$[g(\rho_b) - g(\rho_a) - \lambda\Delta\rho]\cos(\delta/2) = -\varepsilon[P(q_b) - P(q_a)] \cdot e((\varphi_a + \varphi_b)/2). \tag{4.9}$$

Since $g' \to 3$, $\lambda \to 0$, and $DP = O(\sigma)$, this proves the radial bound.

A second complete pair would also have vanishing total mass, which is impossible. The exterior angular equations therefore give separation h. The pair's weighted T-average, divided by ν after internal cancellation, gives the probe equation. Finally divide (4.6) by δ: its radial contribution tends to zero, its angular derivative tends to D, and its right side is $\nu/d^3 + o(1)$. Lemma 2.3 gives $D > 0$. □

4.3 A complete triple

Proposition 4.3. *Let E be a complete triple with exterior label p, $\nu = \sum_{i\in E} \mu_i$, $\sigma = 1 - \nu$, diameter d, and angular width W. It is an endpoint block and*

$$\begin{gathered} \nu \to 0, \quad \sigma \to 1, \quad |\varphi_p - \varphi_i| \to h \quad (i \in E), \\ \max_{i,j\in E} |\rho_i - \rho_j| = o(d), \qquad W/d \to 1, \qquad 0 < C^{-1} \le \nu/d^3 \le C\sigma. \end{gathered} \tag{4.10}$$

Proof. It suffices to treat $E = \{1, 2, 3\}$, $p = 4$. Put

$$J(\rho, v) = \rho\rho_p \sin(\varphi_p - v)(|\rho e(v) - q_p|^{-3} - c_\varepsilon), \qquad J_i = J(\rho_i, \varphi_i).$$

Summing the weighted triple equations and cancelling $\sigma\nu$ gives $f(\varphi_p^0 - \psi) = 0$ at the limiting center ψ. Separation and (4.1) force $\varphi_p^0 - \psi = h$. On the triple's polar rectangle,

$$J_\rho \to -3\sqrt{3}/4, \qquad J_v \to D_0 := 9/4. \tag{4.11}$$

Expanding (4.4) against the exterior gives $\rho_j - \rho_i \ge -C(\varphi_j - \varphi_i)$ for adjacent, hence all ordered, triple pairs. The first and last angular equations give $J_1 < 0 < J_3$. If $\rho_3 \ge \rho_1$, integration of (4.11) yields

$$0 < J_3 - J_1 \le -(\rho_3 - \rho_1)/C + CW.$$

The opposite case uses the support bound. All intermediate radii lie between $\rho_1 - CW$ and $\rho_3 + CW$, so the radial spread is $O(W)$ and $d \asymp W$.

For internal pairs, $k_{ij} = \rho_i\rho_j s(\varphi_j - \varphi_i)(r_{ij}^{-3} - c_\varepsilon) \ge C^{-1}d^{-3}$. Subtracting the first and last angular equations, each internal mass multiplies angular distances summing to W. Hence

$$\nu W/d^3 \le C\sigma(J_3 - J_1) \le C\sigma W.$$

Thus $\nu/d^3 \le C\sigma$, $\nu \to 0$, and $\sigma \to 1$.

Along any further subsequence, extract limits of $(q_i - q_1)/d$ and group equal limits. There are at least two groups. Average (3.3) within each group with its current normalized masses. Internal forces cancel; other groups contribute at most $C\nu/d^2 = O(d)$ before multiplication by

ε. Each group has diameter $o(d)$, while the exterior field has bounded derivative. Subtracting the averages gives, for representatives,

$$F_0(q_i) - F_0(q_j) = o(d) + O(\varepsilon d).$$

Since $DF_0 \to 3e(\psi)e(\psi)^T$, radial differences between groups are $o(d)$, and the same holds within groups. Equation (4.3) now gives $W/d \to 1$.

If $\nu/d^3 \to 0$, repeat these normalized averages in the angular equations. Cross-group interactions are $O(\nu/d^2) = o(d)$, and each averaged J differs from its representative value by $o(d)$. Subtraction, (4.11), and the radial estimate give $D_0(\varphi_i - \varphi_j) = o(d)$, contradicting distinct scaled groups. This proves the lower bound. Reflection gives the other endpoint block. □

4.4 A smaller pair inside a triple

Proposition 4.4. *Suppose the shortest internal distance s in Proposition 4.3 satisfies $s/d \to 0$. Choose fixed shortest-pair labels $a < b$, and let c be the third label. Put $\xi = \mu_a + \mu_b$, $\delta = \varphi_b - \varphi_a$, and $\Delta\rho = \rho_b - \rho_a$. The pair is adjacent and*

$$\xi/\nu \to 0, \qquad \nu/d^3 \to D_0, \qquad \xi/s^3 \to 3D_0, \qquad \delta/s \to 1, \qquad |\Delta\rho| \le C\varepsilon s. \tag{4.12}$$

These limits do not require convergence of μ_a/ξ.

Proof. The shortest pair is adjacent: otherwise $W = \delta \le Cs = o(d)$. Let $\zeta = 1$ for a left triple and -1 for a right triple, and $\omega = \operatorname{sign}(\varphi_c - \varphi_a)$. The distances from the pair to c are $d(1 + o(1))$; their angular differences divided by d tend to ω, and their radial differences are $o(d)$. These estimates hold throughout the pair's segment and polar rectangle.

Define P, H as in the pair proof, now summing over $j = c, p$, and put $D_j = |\rho e(v) - q_j|$, $\Delta_j = \varphi_j - v$. Since $\mu_c/d^3 \le C$, $DP = O(1)$. Differentiation gives

$$\begin{aligned} H_\rho &= -3 \sum_{j=c,p} \mu_j \rho_j \sin\Delta_j (\rho - \rho_j \cos\Delta_j) D_j^{-5}, \\ H_v &= \sum_{j=c,p} \mu_j \rho_j [-\cos\Delta_j (D_j^{-3} - \rho_j^{-3}) + 3\rho\rho_j \sin^2\Delta_j D_j^{-5}]. \end{aligned} \tag{4.13}$$

The near source contributes $o(1)$ to H_ρ and $2\mu_c/d^3 + o(1)$ to H_v; the exterior contributes $-\zeta 3\sqrt{3}/4$ and D_0. Thus

$$\zeta H_\rho \le -C^{-1}, \qquad |H_\rho| + |H_v| \le C, \qquad H_v = D_0 + 2\mu_c/d^3 + o(1). \tag{4.14}$$

The vector difference (4.8), with d, ν replaced by s, ξ, again has only radial or tangential limits. In the radial case, support against p forces $\operatorname{sign}(\Delta\rho) = \zeta$. Since $\delta = o(s)$ and $|\Delta\rho| \sim s$, (4.14) makes the left side of (4.6) negative, contradicting its positive right side. Hence $\lambda \to 0$, $\delta/s \to 1$, and $\Delta\rho/s \to 0$. The angular difference gives $\xi/s^3 \le C$, whence

$$\frac{\xi}{\nu} = \frac{\xi/s^3}{\nu/d^3} \left(\frac{s}{d}\right)^3 \to 0.$$

The midpoint identity (4.9), with $DP = O(1)$, gives $|\Delta\rho| \le C\varepsilon s$.

Average the pair's T equations with weights $\mu_a/\xi, \mu_b/\xi$ and subtract the equation of c. Let K be the averaged variational kernel from the pair to c, and J_0 its averaged exterior field. Internal cancellation gives exactly

$$0 = \nu K + \sigma(J_0 - J_c). \tag{4.15}$$

The triple estimates imply $K = \omega d^{-2}(1 + o(1))$ and $J_0 - J_c = -D_0\omega d + o(d)$. Division by ωd gives $\nu/d^3 \to D_0$, hence $\mu_c/d^3 \to D_0$. Now $H_v \to 3D_0$ in (4.14); dividing the pair angular difference by δ, with $\Delta\rho/\delta = O(\varepsilon)$, gives $\xi/s^3 \to 3D_0$. All averages use current normalized masses, so the conclusions hold on the selected fixed-label sequence. □

Every sequence therefore has, after extraction, one of four types: no satellite collision, a complete pair, a triple with comparable internal distances, or a triple containing a smaller pair. The span bound excludes a four-member cluster, and Proposition 4.2 excludes two pairs.

5 Regularized solution charts

We use the residuals (3.5)–(3.6), changing only coordinates and rows. Put

$$\chi(t, z) = \tfrac{1}{2}tz^2 s(tz/2)^2, \qquad 1 - \cos(tz) = t\chi(t, z). \tag{5.1}$$

The extensions to signed scales below are analytic extensions of the positive-scale equations. For a smooth target field K,

$$\frac{K(\rho + tu, v + tz) - K(\rho, v)}{t} = \int_0^1 (uK_\rho + zK_v)(\rho + \lambda tu, v + \lambda tz)\,\mathrm{d}\lambda. \tag{5.2}$$

Here source positions and the common vector X are fixed in the target difference; subsequent differentiation includes their dependence on the variables.

One formula covers the internal terms of both single-scale charts. If $\mu_i = t^3\alpha_i$, $\rho_i = R + tw_i$, $\varphi_i = \psi + tz_i$, set $a_{ij} = w_i - w_j$, $b_{ij} = z_i - z_j$ and

$$W_{ij}^2 = a_{ij}^2 + \rho_i\rho_j b_{ij}^2 s(tb_{ij}/2)^2, \qquad r_{ij} = tW_{ij} \quad (t > 0).$$

The internal contributions to R_i, T_i are, respectively,

$$\begin{aligned} &- \varepsilon t \sum_{j\neq i} \alpha_j [a_{ij} + \rho_j \chi(t, b_{ij})] W_{ij}^{-3}, \\ &- t \sum_{j\neq i} \alpha_j \rho_i \rho_j b_{ij} s(tb_{ij}) (W_{ij}^{-3} - t^3 c_\varepsilon). \end{aligned} \tag{5.3}$$

Thus their divided differences are smooth wherever $W_{ij} > 0$. Their weighted angular sum vanishes, and their weighted radial sum is

$$-\varepsilon \sum_{i<j} \alpha_i\alpha_j(\rho_i + \rho_j)[1 - \cos(tb_{ij})]W_{ij}^{-3}.$$

All exterior terms are smooth target fields and use (5.2).

5.1 No collision

At a distinct limiting root with mass μ^0 in the closed simplex, choose p with $\mu_p^0 \geq 1/4$, fix $\varphi_p = 0$ and omit T_p. The seven residuals are smooth for nearby signed masses of sum one. Their derivative at $\varepsilon = 0$ has the form

$$\begin{pmatrix} 3I_4 & 0 \\ * & K \end{pmatrix},$$

where K is the reduced derivative from Proposition 2.4. Lemma 3.2 gives a unique root on an independent product in ε and the mass coordinates. Convexity persists since all cyclic triple areas have positive limits; Lemma 3.1 gives nondegeneracy at positive masses. Convergence places every sequence with these limiting data in the root neighborhood. The omitted angular equation follows from (3.7).

5.2 A complete pair

Fix limiting exterior angles $0, h$, exterior mass fraction p_0, and pair angle ψ_0 as in Proposition 4.2. Let $D > 0$ be the probe derivative there and $z_0 = D^{-1/3}$. For independent parameters $(\varepsilon, t, p, \alpha)$, put $\beta = 1 - \alpha$ and

$$\mu_a = t^3\alpha, \quad \mu_b = t^3\beta, \quad \mu_c = (1-t^3)p, \quad \mu_d = (1-t^3)(1-p), \tag{5.4}$$

$$\varphi_c = 0, \quad \varphi_d = \omega, \quad \rho_a = R - \beta t u, \quad \rho_b = R + \alpha t u, \quad \varphi_a = \psi - \beta t z, \quad \varphi_b = \psi + \alpha t z. \tag{5.5}$$

Use variables $(\rho_c, \rho_d, R, u, \omega, \psi, z)$. For either row family $L = R, T$, put $A_L = \alpha L_a + \beta L_b$, $B_L = (L_b - L_a)/t$, and retain

$$F = (R_c, R_d, A_R, B_R, T_o, A_T, B_T), \qquad o = \begin{cases} d, & p_0 \geq 1/2, \\ c, & p_0 < 1/2. \end{cases}$$

The omitted exterior row has positive mass bounded away from zero. Both coordinate changes have determinant t; the inverse row formulas $L_a = A_L - \beta t B_L$, $L_b = A_L + \alpha t B_L$ require no division by a mass.

Here

$$r_{ab}^2 = t^2 W^2, \qquad W^2 = u^2 + \rho_a \rho_b z^2 s(tz/2)^2. \tag{5.6}$$

Equation (5.3) and (5.2) extend every row smoothly near $W = z_0 > 0$, uniformly for α in an interval containing $[0, 1]$. At $\varepsilon = t = 0$ the radial rows are $g(\rho_c), g(\rho_d), g(R), g'(R)u$. With $(p_c, p_d) = (p, 1-p)$ and

$$H(\rho, v) = \rho \sum_{j=c,d} p_j \rho_j \sin(\varphi_j - v)(|\rho e(v) - q_j|^{-3} - 1),$$

the cluster angular rows become

$$A_T = H(R, \psi), \qquad B_T = H_\rho u + H_v z - \frac{R^2 z}{(u^2 + R^2 z^2)^{3/2}}. \tag{5.7}$$

At $y^0 = (1, 1, 1, 0, h, \psi_0, z_0)$ the radial block is $(3I_4, 0)$ and the angular block is lower triangular, with diagonal

$$(a_0, D, 3D), \qquad a_0 = \begin{cases} D_0 p_0, & o = d, \\ -D_0(1-p_0), & o = c. \end{cases} \tag{5.8}$$

Since $|a_0| \geq 9/8$, the derivative is invertible. The base and derivative are independent of α, so Lemma 3.2 gives a common root on an independent (ε, t, p) product, uniformly in $\alpha \in [0, 1]$. At $\varepsilon = 0$ row recovery gives unit radii and $u = 0$; hence $u = O(\varepsilon)$ uniformly.

Convexity follows from the areas containing the pair. Set $q_0 = 0$ and $S_j = \det(q_b - q_a, q_j - q_a)$. At the base point, their smooth quotients satisfy

$$(S_0/t)|_0 = z_0, \qquad (S_j/t)|_0 = z_0[1 - \cos(\varphi_j^0 - \psi_0)] > 0 \quad (j = c, d). \tag{5.9}$$

All other cyclic triple areas have positive limits. The invertible row and coordinate changes and Lemma 3.1 give nondegeneracy. Finally, Proposition 4.2 gives $t = \nu^{1/3}$, $z = \delta/t \to z_0$, $u = (\rho_b - \rho_a)/t \to 0$, and, after extraction, $p \to p_0$. The actual variables therefore enter the common root ball, without convergence of $\alpha = \mu_a/\nu$.

5.3 A triple with comparable distances

Label the triple $1,2,3$ in increasing angular order and its exterior p. Its limiting angle in the gauge $\varphi_p = 0$ is τh, $\tau = \pm 1$. Consider

$$\mathcal{G}_i(z;\alpha) = D_0 z_i - \sum_{j\neq i} \alpha_j \frac{z_i - z_j}{|z_i - z_j|^3} = 0, \qquad z_1 < z_2 < z_3, \quad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1. \tag{5.10}$$

For positive weights these are the divided gradient equations of the proper, strictly convex function

$$\mathcal{V}(z) = \frac{D_0}{2} \sum_i \alpha_i z_i^2 + \sum_{i<j} \frac{\alpha_i \alpha_j}{|z_i - z_j|},$$

so there is exactly one root and its weighted center is zero. For nonnegative weights the divided derivative $J = D_z \mathcal{G}$ has entries

$$J_{ii} = D_0 + 2\sum_{j\neq i} \frac{\alpha_j}{|z_i - z_j|^3}, \qquad J_{ij} = -\frac{2\alpha_j}{|z_i - z_j|^3} \quad (i \neq j). \tag{5.11}$$

Strict row diagonal dominance gives $\|Jv\|_\infty \geq D_0 \|v\|_\infty$. The integral of J along any ordered segment has the same property, proving uniqueness also for nonnegative weights. Boundary existence is needed only at limits obtained below.

Set

$$L = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \qquad B_\alpha = \begin{pmatrix} -(\alpha_2 + \alpha_3) & -\alpha_3 \\ \alpha_1 & -\alpha_3 \\ \alpha_1 & \alpha_1 + \alpha_2 \end{pmatrix}. \tag{5.12}$$

Then $LB_\alpha = I_2$, $\alpha^T B_\alpha = 0$. Fix a distinct root z^0 at α^0, put $v^0 = Lz^0$, and use

$$\begin{gathered} \mu_i = t^3 \alpha_i, \quad \mu_p = 1 - t^3, \qquad w = B_\alpha u, \quad z = B_\alpha v, \\ \rho_p = r, \quad \varphi_p = 0, \quad \rho_i = R + t w_i, \quad \varphi_i = \psi + t z_i. \end{gathered} \tag{5.13}$$

For $P = (P_1, P_2, P_3)^T$, either R or T, retain $A_P = \alpha^T P$ and $B_P = LP/t$, together with R_p. The inverse $P = A_P \mathbf{1} + t B_\alpha B_P$ retains every probe row, including when $\alpha_i = 0$; T_p follows since $\mu_p > 0$. Equation (5.3) proves smoothness on an independent (ε, t, α) neighborhood: all W_{ij} are positive at z^0.

At $\varepsilon = t = 0$ the radial rows are $g(r), g(R), g'(R)u_1, g'(R)u_2$. Put

$$\mathcal{H}(\rho, v; r) = -\rho r \sin v \big((\rho^2 + r^2 - 2\rho r \cos v)^{-3/2} - 1\big). \tag{5.14}$$

The angular rows are $\mathcal{H}(R, \psi; r)$ and LM, where

$$M_i = \mathcal{H}_\rho w_i + \mathcal{H}_v z_i - \sum_{j\neq i} \frac{\alpha_j R^2 (z_i - z_j)}{[(w_i - w_j)^2 + R^2 (z_i - z_j)^2]^{3/2}}.$$

At $(r, R, u, \psi) = (1, 1, 0, \tau h)$ the last rows are $L\mathcal{G}$. Since $\alpha^T \mathcal{G} = D_0 \alpha^T z = 0$, they recover all of (5.10). At $y^0 = (1, 1, 0, 0, \tau h, v_1^0, v_2^0)$ the radial block is $(3I_4, 0)$ and the angular block has diagonal blocks D_0 and

$$K = LJB_\alpha. \tag{5.15}$$

If $K\xi = 0$, then $JB_\alpha \xi = c_1 \mathbf{1}$. The identity $\alpha^T J = D_0 \alpha^T$ gives $c_1 = 0$, hence $\xi = 0$. Thus Lemma 3.2 applies. At $\varepsilon = 0$ row recovery gives $r = R = 1, u = 0$, including $t = 0$ by continuity, so $|r-1| + |R-1| + |u| = O(\varepsilon)$. Nondegeneracy follows from Lemma 3.1; convexity is proved in Section 6.

For capture, Proposition 4.3 gives $t = \nu^{1/3} \asymp d$, $u \to 0$ and bounded positive gaps v. Extract $\alpha \to \alpha^0$, $v \to v^0$ with both gaps positive. The smooth difference rows limit to $L\mathcal{G} = 0$, hence (5.10). The resulting boundary root supplies the chart, and local uniqueness captures the sequence.

5.4 A nested triple

Let $a < b$ be the smaller pair, c the third cluster member and p the exterior. Let $\tau = \pm 1$ specify the cluster side relative to p, and $\omega = \pm 1$ the side of c relative to the pair. Put $\beta = 1 - \alpha$, $\gamma = \eta^3$, and

$$\mu_a = t^3\gamma\alpha, \quad \mu_b = t^3\gamma\beta, \quad \mu_c = t^3(1-\gamma), \quad \mu_p = 1 - t^3. \tag{5.16}$$

Use $y = (\rho_p, R, u_1, u_2, \psi, z_1, z_2)$ and

$$\begin{aligned} S &= R - (1-\gamma)tu_1, & \psi_0 &= \psi - (1-\gamma)tz_1, \\ \rho_a &= S - \beta t\eta u_2, & \varphi_a &= \psi_0 - \beta t\eta z_2, \\ \rho_b &= S + \alpha t\eta u_2, & \varphi_b &= \psi_0 + \alpha t\eta z_2, \\ \rho_c &= R + \gamma t u_1, & \varphi_c &= \psi + \gamma t z_1, \qquad \varphi_p = 0. \end{aligned} \tag{5.17}$$

For either row family $P = R, T$, define

$$P_0 = \alpha P_a + \beta P_b, \quad A_P = \gamma P_0 + (1-\gamma)P_c, \quad B_P = (P_c - P_0)/t, \quad C_P = (P_b - P_a)/(t\eta). \tag{5.18}$$

Retain $(R_p, A_R, B_R, C_R, A_T, B_T, C_T)$. The inverse is

$$\begin{aligned} P_0 &= A_P - (1-\gamma)tB_P, & P_c &= A_P + \gamma t B_P, \\ P_a &= P_0 - \beta t\eta C_P, & P_b &= P_0 + \alpha t\eta C_P. \end{aligned} \tag{5.19}$$

The coordinate determinant has absolute value $t^4\eta^2$, and $(1-t^3)T_p + t^3 A_T = 0$ recovers the exterior row. No row is lost at $\alpha = 0, 1$.

Smoothness in the independent scales. Put $P_a = u_1 + \beta\eta u_2$, $P_b = u_1 - \alpha\eta u_2$, $Q_a = z_1 + \beta\eta z_2$, $Q_b = z_1 - \alpha\eta z_2$. Then $r_{ab} = t\eta W$, $r_{ic} = tV_i$ for $i = a, b$, where

$$W^2 = u_2^2 + \rho_a\rho_b z_2^2 s(t\eta z_2/2)^2, \qquad V_i^2 = P_i^2 + \rho_i\rho_c Q_i^2 s(tQ_i/2)^2. \tag{5.20}$$

Let H, K be the exterior angular and radial fields,

$$\begin{aligned} H(\rho, v) &= -(1-t^3)\rho\rho_p \sin v(|\rho e(v) - \rho_p e(0)|^{-3} - c_\varepsilon), \\ K(\rho, v) &= g(\rho) - \varepsilon(1-t^3)\frac{\rho - \rho_p\cos v}{|\rho e(v) - \rho_p e(0)|^3} - \varepsilon c_\varepsilon X \cdot e(v). \end{aligned}$$

For target values write $H_i = H(\rho_i, \varphi_i)$, and similarly for K; index 0 denotes the average with weights α, β. Define

$$L_i = \rho_i\rho_c Q_i s(tQ_i)(V_i^{-3} - t^3 c_\varepsilon), \qquad M = \rho_a\rho_b z_2 s(t\eta z_2)(W^{-3} - t^3\eta^3 c_\varepsilon).$$

Exact substitution gives

$$\begin{aligned} A_T &= \gamma H_0 + (1-\gamma)H_c, \\ B_T &= (H_c - H_0)/t - L_0, \\ C_T &= (H_b - H_a)/(t\eta) + (1-\gamma)(L_b - L_a)/\eta - M. \end{aligned} \tag{5.21}$$

The field differences use (5.2). For the remaining quotient define

$$\mathcal{L}(r, P, Q) = r\rho_c Q s(tQ)[(P^2 + r\rho_c Q^2 s(tQ/2)^2)^{-3/2} - t^3 c_\varepsilon].$$

Along $(r,P,Q)=(S+t\eta\lambda u_2,u_1-\eta\lambda u_2,z_1-\eta\lambda z_2)$,

$$\frac{L_b-L_a}{\eta}=\int_{-\beta}^{\alpha}(tu_2\mathcal{L}_r-u_2\mathcal{L}_P-z_2\mathcal{L}_Q)(r,P,Q)\,\mathrm{d}\lambda. \tag{5.22}$$

The substitution $\lambda+\beta$ fixes the interval to $[0,1]$.

For the radial terms put

$$\begin{aligned} N_i&=(P_i-\rho_c\chi(t,Q_i))V_i^{-3}, \qquad O_i=(P_i+\rho_i\chi(t,Q_i))V_i^{-3},\\ N&=(u_2-\rho_b\chi(t\eta,z_2))W^{-3}, \quad O=(u_2+\rho_a\chi(t\eta,z_2))W^{-3}, \end{aligned}$$

and $J=-\varepsilon t\eta\alpha\beta(\rho_a+\rho_b)\chi(t\eta,z_2)W^{-3}$. The exact rows are

$$\begin{aligned} A_R&=\gamma K_0+(1-\gamma)K_c+\varepsilon t\gamma(1-\gamma)(N_0-O_0)+\gamma J,\\ B_R&=(K_c-K_0)/t-\varepsilon[\gamma O_0+(1-\gamma)N_0]-J/t,\\ C_R&=(K_b-K_a)/(t\eta)+\varepsilon(1-\gamma)(N_b-N_a)/\eta-\varepsilon(\alpha O+\beta N). \end{aligned} \tag{5.23}$$

Here J/t is explicitly smooth. The quotient $(N_b-N_a)/\eta$ has formula (5.22) with

$$\mathcal{N}(r,P,Q)=(P-\rho_c\chi(t,Q))(P^2+r\rho_cQ^2s(tQ/2)^2)^{-3/2}.$$

The exterior row has no internal denominator. Positive W,V_a,V_b therefore give joint smoothness in $(\varepsilon,t,\eta,\alpha)$, including all scale faces and both zero-mass endpoints.

Boundary equations and continuation. At $\varepsilon=t=\eta=0$ the radial rows are $g(\rho_p),g(R),g'(R)u_1,g'(R)u_2$. Put

$$L_0=\frac{R^2z_1}{(u_1^2+R^2z_1^2)^{3/2}}, \qquad M_0=\frac{R^2z_2}{(u_2^2+R^2z_2^2)^{3/2}}.$$

With $\mathcal{H}$ from (5.14), the angular rows are

$$\begin{aligned} A_T&=\mathcal{H}(R,\psi;\rho_p),\\ B_T&=\mathcal{H}_\rho u_1+\mathcal{H}_v z_1-L_0,\\ C_T&=\mathcal{H}_\rho u_2+\mathcal{H}_v z_2-u_2\partial_{u_1}L_0-z_2\partial_{z_1}L_0-M_0. \end{aligned} \tag{5.24}$$

They vanish at

$$y^0=(1,1,0,0,\tau h,Z_0,z_0), \qquad Z_0=\omega D_0^{-1/3}, \quad z_0=(3D_0)^{-1/3}. \tag{5.25}$$

The radial block is $(3I_4,0)$ and the angular block is

$$\begin{pmatrix} D_0 & 0 & 0\\ CZ_0 & 3D_0 & 0\\ Cz_0 & -6D_0z_0/Z_0 & 9D_0 \end{pmatrix}, \qquad C=-\tfrac{27}{4}\sin(\tau h). \tag{5.26}$$

Indeed $\partial_{z_1}L_0=-2D_0$, $\partial_{z_1}^2L_0=6D_0/Z_0$ and $\partial_{z_2}M_0=-6D_0$ at the base. Hence $\det D_yF=2187D_0^3\neq 0$. The base, derivative and positive denominator bounds are independent of α. Lemma 3.2 supplies one root and a bounded inverse on an independent (ε,t,η) product, uniformly for $\alpha\in[0,1]$. At $\varepsilon=0$ row recovery gives unit radii and $u_1=u_2=0$; continuity covers the scale faces. Thus $u_j=O(\varepsilon)$ uniformly. At positive parameters Lemma 3.1 gives nondegeneracy.

For capture, take $t = \nu^{1/3}$, $\eta = (\xi/\nu)^{1/3}$, $\alpha = \mu_a/\xi$. Proposition 4.4 gives

$$t, \eta \to 0, \quad d/t \to D_0^{-1/3}, \quad z_2 = \delta/(t\eta) \to z_0, \quad u_2 = (\rho_b - \rho_a)/(t\eta) = O(\varepsilon) \to 0.$$

The larger-scale radial estimate gives $u_1 \to 0$. The two signed angular differences from the pair to c, divided by d, tend to ω, so averaging gives $z_1 \to Z_0$. Also $\rho_p, R \to 1$ and $\psi \to \tau h$. The variables enter the common root ball without convergence of α. The four possibilities are

E	$\{a, b\}$	p	(τ, ω)
$\{1, 2, 3\}$	$\{1, 2\}$	4	$(-1, +1)$
$\{1, 2, 3\}$	$\{2, 3\}$	4	$(-1, -1)$
$\{2, 3, 4\}$	$\{2, 3\}$	1	$(+1, +1)$
$\{2, 3, 4\}$	$\{3, 4\}$	1	$(+1, -1)$.

Extracting fixed labels selects one chart. Section 6 proves strict convexity and identifies its fixed-mass endpoint at $\varepsilon = 0$.

6 Strict convexity and fixed-mass endpoints

The triple charts require exact collinearity at $t = 0$ to preserve convexity on an independent parameter product.

6.1 Collinearity at zero cluster scale

Set $a = (1 + \varepsilon)^{1/3}$, $e_0 = e(\tau h)$, $w_0 = e(0) - e_0$ and $q^0 = ae_0$. At $t = 0$, the exterior is $q_p = ae(0)$ and the probe field is

$$F(q) = q - \frac{q}{|q|^3} + \varepsilon \frac{q_p - q}{|q_p - q|^3} - \varepsilon \frac{q_p}{a^3}.$$

The equilateral background gives $F(q^0) = 0$. In the basis $(e_0, e_0^\perp)$ its derivative is

$$B = \frac{3}{1+\varepsilon}(e_0 e_0^T + \varepsilon w_0 w_0^T) = \frac{3}{1+\varepsilon} \begin{pmatrix} 1 + \varepsilon/4 & \tau\sqrt{3}\varepsilon/4 \\ \tau\sqrt{3}\varepsilon/4 & 3\varepsilon/4 \end{pmatrix}. \tag{6.1}$$

Its trace is 3 and its smaller eigenvalue is

$$\lambda = \varepsilon d, \qquad d = \frac{9}{2(1+\varepsilon)^2(1 + \sqrt{1 - 3\varepsilon/(1+\varepsilon)^2})}, \qquad d(0) = D_0. \tag{6.2}$$

All quantities below depend smoothly on small ε:

$$k = -\frac{B_{12}}{B_{11} - \lambda}, \qquad n = \frac{ke_0 + e_0^\perp}{\sqrt{1 + k^2}}, \qquad l = (D_0/d)^{1/3}, \qquad b = \frac{l}{a\sqrt{1 + k^2}}.$$

Here $Bn = \lambda n$, $k(0) = 0$ and $l(0) = b(0) = 1$. If distinct scalar positions s_i satisfy

$$D_0 s_i - \sum_{j \neq i} \omega_j \frac{s_i - s_j}{|s_i - s_j|^3} = 0, \qquad \sum_i \omega_i s_i = 0, \quad \sum_i \omega_i = 1, \tag{6.3}$$

then $b_i = l s_i n$ satisfy

$$B b_i - \varepsilon \sum_{j \neq i} \omega_j \frac{b_i - b_j}{|b_i - b_j|^3} = 0, \tag{6.4}$$

since $dl = D_0 l^{-2}$.

For fixed $\varepsilon \neq 0$ and, in the nested chart, $\eta > 0$, substitute $q_i(t) = q^0 + tb_i + O(t^2)$ and $\mu_i = t^3\omega_i$ in the Cartesian equations. Their first-order terms are t times the left side of (6.4); the exterior mass and reaction vector change only at order t^3. The boundary difference rows are its radial projections and a/ε times its tangential projections. Basis derivatives contribute nothing because $F(q^0) = 0$; the center and exterior rows vanish on the background. Thus (6.4) solves these boundary equations. Their joint smooth extensions from Section 5 also give the identities at zero parameters.

For the regular chart take $s_i = z_i(0, 0, \alpha)$ from its own root. Weighted row recovery gives (6.3). The resulting boundary root is

$$r = R = a, \qquad \psi = \tau h, \qquad v = bv(0, 0, \alpha), \qquad u = akv. \tag{6.5}$$

The factor a^{-1} in b converts tangential to angular displacement. This candidate stays in the chart's common neighborhood, so uniqueness identifies it with the root at $t = 0$. In particular,

$$w_i = akz_i \qquad (t = 0). \tag{6.6}$$

For the nested chart, its root at $\varepsilon = t = 0$, with $\eta > 0$ fixed, gives

$$s_a = -(1-\gamma)z_1 - \beta\eta z_2, \quad s_b = -(1-\gamma)z_1 + \alpha\eta z_2, \quad s_c = \gamma z_1,$$

with weights $(\gamma\alpha, \gamma\beta, 1-\gamma)$. The angular difference rows make the three scalar residuals equal, and their weighted sum is zero. Hence (6.3) holds, including $\alpha = 0, 1$. Set $\rho_p = R = a$, $\psi = \tau h$, multiply both z_j by b, and set $u_j = akz_j$. This boundary root is smooth in η, α and stays in the common neighborhood. Uniqueness and continuity as $\eta \downarrow 0$ yield

$$u_2 z_1 - u_1 z_2 = 0 \qquad (t = 0) \tag{6.7}$$

for every small ε and all $\alpha \in [0, 1]$.

6.2 Signed areas

For $A < B < C < A + \pi$,

$$\det(e(B) - e(A), e(C) - e(A)) = 4 \sin\frac{B-A}{2} \sin\frac{C-B}{2} \sin\frac{C-A}{2} > 0. \tag{6.8}$$

With the origin as third vertex, the corresponding area is $\sin(B - A) > 0$.

In the regular chart, line-segment integration gives smooth divided edges

$$q_j - q_i = tV_{ij}, \qquad V_{ij}|_{t=0} = (w_j - w_i)e_0 + R(z_j - z_i)e_0^{\perp}.$$

Evaluate $B_1 = \det(V_{12}, V_{13})$ along the smooth chart root. Equation (6.6) gives $B_1(\varepsilon, 0, \alpha) = 0$, so

$$\det(q_2 - q_1, q_3 - q_1) = t^3 C_1, \qquad C_1(\varepsilon, t, \alpha) = \int_0^1 \partial_t B_1(\varepsilon, st, \alpha)\, \mathrm{d}s. \tag{6.9}$$

The derivative includes the root's dependence on t. At $\varepsilon = 0$ every radius is one; writing $v^0 = v(0, 0, \alpha)$, (6.8) gives

$$C_1(0, 0, \alpha) = \tfrac{1}{2} v_1^0 v_2^0 (v_1^0 + v_2^0) > 0.$$

Thus C_1 stays positive on a smaller product. Areas containing both exterior vertices have positive equilateral limits. Those containing two cluster members and one exterior vertex

have a factor t; at the base their quotients are $z_j - z_i$ for the origin and $(z_j - z_i)(1 - \cos h)$ for p, with cyclic order understood. They are positive.

In the nested chart write

$$q_b - q_a = t\eta V_{ab}, \qquad q_c - q_a = tV_{ac}, \qquad B_2 = \det(V_{ab}, V_{ac})$$

along its root. At $t = 0$,

$$B_2 = R\{u_2(z_1 + \beta\eta z_2) - z_2(u_1 + \beta\eta u_2)\} = R(u_2 z_1 - u_1 z_2) = 0.$$

Hence

$$\det(q_b - q_a, q_c - q_a) = t^3\eta C_2, \qquad C_2 = \int_0^1 \partial_t B_2(\varepsilon, st, \eta, \alpha)\,\mathrm{d}s. \tag{6.10}$$

Again the derivative is along the root. Equation (6.8) gives

$$C_2(0, 0, 0, \alpha) = \tfrac{1}{2} z_0 Z_0^2 > 0.$$

This is independent of α and positive for either sign of Z_0; (c, a, b) and (a, b, c) are cyclic permutations. The other area quotients at $\varepsilon = t = \eta = 0$, in cyclic order, are as follows:

vertices	factor	limiting quotient
$0, p$, one cluster member	1	$\sin h$
$0, a, b$	$t\eta$	z_0
p, a, b	$t\eta$	$z_0(1 - \cos h)$
$0, c$, one pair member	t	$\lvert Z_0\rvert$
p, c, one pair member	t	$\lvert Z_0\rvert(1 - \cos h)$

Continuity preserves all ten cyclic triple areas on a common product, uniformly for $\alpha \in [0, 1]$. Thus every consecutive edge has the other vertices strictly on its left, proving strict convexity in both triple charts.

Proposition 6.1. *After restriction to smaller independent parameter products, all four chart types capture their actual sequences. Their roots with positive satellite masses and $\varepsilon > 0$ are strictly convex in the prescribed order and have nondegenerate shape Hessian. Fixing any positive mass vector represented by a chart and varying only ε gives a smooth branch on $[0, \bar{\varepsilon})$. At zero it has unit radii and distinct coorbital angles in the same open semicircle, for that same mass vector. For a captured sequence, $\bar{\varepsilon} > 0$ is independent of its index after extraction into one chart.*

Proof. Smoothness, capture and nondegeneracy follow from Section 5; convexity follows from the area factors above and the pair and noncollision estimates there. With positive mass coordinates fixed, the scale parameters t and, when present, η are positive. The row transformations remain invertible at $\varepsilon = 0$: radial recovery gives $g(\rho_i) = 0$, hence $\rho_i = 1$, and the weighted angular identity recovers the omitted row. The pair gap tz, regular gaps tv_i, and nested gaps $t\eta z_2, tQ_a, tQ_b$ remain nonzero in their prescribed order. Exterior separations and span $< \pi$ persist. Consequently the endpoint is a noncollision coorbital root for the fixed positive masses. Each parameter restriction is made once about the selected base; freezing the mass coordinates therefore leaves a common ε interval for the captured sequence. □

7 Connection of all solutions and the common threshold

Let $\mathcal{S}(\varepsilon, \mu)$ consist of the strictly convex multiplier-one representatives in standard directed order,

$$y = (\rho_1, \rho_2, \rho_3, \rho_4, \varphi_2, \varphi_3, \varphi_4), \qquad \varphi_1 = 0.$$

Each allowed similarity class has exactly one such representative.

Proposition 7.1. *If $y_n \in \mathcal{S}(\varepsilon_n, \mu^{(n)})$ and $\varepsilon_n \to 0$, then a subsequence admits smooth branches $\Gamma_n : [0, \bar{\varepsilon}) \to \mathbb{R}^7$, with $\bar{\varepsilon} > 0$ independent of n, such that $\Gamma_n(\varepsilon_n) = y_n$ and*

$$\Gamma_n(s) \in \mathcal{S}(s, \mu^{(n)}) \qquad (0 < s < \bar{\varepsilon}).$$

Their shape Hessians are nondegenerate for $s > 0$, and their zero endpoints are distinct-angle coorbital roots for $\mu^{(n)}$ in the same order.

Proof. Proposition 4.1 gives unit-radius limits and bounds the span away from both zero and π. Extract limits of the angles and masses. Equal-angle blocks are consecutive; a four-member block is impossible, and Proposition 4.2 excludes two pairs. For a triple, extract the shortest-to-largest distance ratio. A positive limit gives the regular chart, after extracting a distinct scalar root; a zero limit gives the nested chart, after fixing the shortest pair. Together with noncollision and one-pair limits, these exhaust the possibilities. The charts capture each case, and Proposition 6.1 gives branches with their mass coordinates fixed on one common ε interval. The pair fractions need not converge because the corresponding charts are uniform on $[0, 1]$. Subtracting the current φ_1 from every angle restores the required gauge smoothly, preserves the equations and nondegeneracy, and returns y_n at ε_n. □

Proof of Theorem 1.1. For each fixed $\mu \in \mathcal{M}$, Proposition 3.3 gives a locally unique convex branch from $y^0(\mu) = (1, 1, 1, 1, \theta_2^0(\mu), \theta_3^0(\mu), \theta_4^0(\mu))$ on $0 \le s < \ell(\mu)$. Theorem 2.2 identifies every endpoint in Proposition 7.1 with this point for its fixed mass vector.

Existence. Otherwise choose $\varepsilon_n \to 0$ and $\mu^{(n)}$ with $\mathcal{S}(\varepsilon_n, \mu^{(n)})$ empty. Choose $0 < s_n < \frac{1}{2}\min\{\varepsilon_n, 1/n, \ell(\mu^{(n)})\}$ and take the local root at s_n. Proposition 7.1 gives branches through these roots on a common interval containing ε_n for large n, a contradiction. Let $\delta_1 > 0$ be a common existence bound.

Uniqueness. Otherwise there are two distinct roots for the same $\mu^{(n)}$ at $\varepsilon_n \to 0$. Apply Proposition 7.1 successively to the two sequences. On a common subsequence their branches Γ_n^1, Γ_n^2 share an interval $[0, \tau)$. Their endpoints coincide, so fixed-mass local uniqueness makes the branches equal near zero. For each sufficiently large n, the set

$$Q_n = \{s \in [0, \tau) : \Gamma_n^1(s) = \Gamma_n^2(s)\}$$

is closed and contains a neighborhood of zero. At every positive point it is open by nondegeneracy, Lemma 3.1 and the implicit function theorem. Connectedness gives $Q_n = [0, \tau)$, contradicting distinctness at ε_n. This yields $\delta_2 > 0$.

Nondegeneracy. A sequence of degenerate roots with $\varepsilon_n \to 0$ would be captured by Proposition 7.1, contradicting nondegeneracy along its positive-parameter branches. Thus there is $\delta_3 > 0$.

Set $\delta_0 = \min\{\delta_1, \delta_2, \delta_3\}$. Simultaneous relabeling of positions and masses takes each directed order to the standard order and maps $\mathcal{M}$ onto itself, so the same bound applies to every order. Undoing the multiplier-one normalization proves the theorem. □

Statements and Declarations

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